

Time Bases Trend Insights & Recommendations

Time Based Trend Insights & Recommendations :-

A.KPI's

1. #_Revenue_Generated

select round(sum(total_amount_to_pay)/10000000,2) as Revenue_CR from fact_orders_enriched;

	Total_Rev_IN_CR
▶	22.22

2. #_PROFIT_Generated

select Round((sum(total_amount_to_pay) - sum(total_Cost_price))/10000000,2) as Profit_CR from fact_orders_enriched;

	Profit_CR
▶	6.70

3. #_PROFIT_Percentage

select Round((sum(total_amount_to_pay) - sum(total_Cost_price))/(sum(total_amount_to_Pay),2)*100 as Profit_PCT from fact_orders_enriched;

	Profit_PCT
▶	30.00

4.Total_Orders_Got_Yet

select Round(count(distinct(order_id))/1000,2) as Total_Orders_Yet_K from fact_orders_enriched ;

	Total_Orders_Yet_K
▶	33.36

B. Year - Week Trend Analysis :-

1. Objective : Analyzing Yearly Revenue & Profit Trend

```
with c1 as (                                -- Added Needed Metrics Year Wise
SELECT
    year,
    ROUND(SUM(total_amount_to_pay) / 10000000, 2) AS Total_Revenue_CR,
    Round((sum( total_amount_to_pay) - sum(total_Cost_price))/10000000,2) as Profit_CR,
    Round((sum( total_amount_to_pay) - sum(total_Cost_price))/(sum(total_amount_to_Pay)),2)*100
as profit_PCT
FROM
    fact_orders_enriched
GROUP BY year
ORDER BY year
),

c2 as (                                     -- Added Vs prev Years Values
select * ,
    Round(((Total_Revenue_CR - lag(Total_Revenue_CR) over( order by
year))*100/lag(Total_Revenue_CR) over( order by year)),2) as Vs_Prev_Year_Revenue,

    Round(((Profit_CR - lag(Profit_CR) over( order by year))*100/lag(Profit_CR) over( order by year)),2) as
Vs_Prev_Year_Profit
from c1
)

select * from c2;
```

Result 

	year	Total_Revenue_CR	Profit_CR	profit_PCT	Vs_Prev_Year_Revenue	Vs_Prev_Year_Profit
2023	2023	11.13	3.35	30.00	NULL	NULL
2024	2024	11.10	3.36	30.00	-0.27	0.30

Insights :-

- Sales Difference between Years Is not that significant 
- Profit PCT has remained Almost Same both Years
- Sales went down slightly by **(- 0.27 %)** From the last Year (YoY)
- Profit_PCT has a slight increase by **(+ 0.30 %)** compared to last Year

2. Objective : Analyzing Quarterly Rev Trend (combined 2 years)

```
with c1 as (                                - - Added Needed Metrics Quarter_wise
SELECT
    quarter_name,
    ROUND(SUM(total_amount_to_pay) / 10000000, 2) AS Total_Revenue_CR
FROM
    fact_orders_enriched
GROUP BY quarter_name
),

c2 as (                                     - - Added contirbution_to_Overall_Rev
select *,
round(Total_Revenue_CR*100/(select sum(Total_amount_to_pay)/10000000 from
fact_orders_enriched),2) as Contirbution_to_overall_Rev
from c1
)
select * from c2;
```

Result 

	quarter_name	Total_Revenue_CR	Contirbution_to_overall_Rev
	Qtr - 1	5.58	25.11
	Qtr - 2	5.61	25.24
	Qtr - 3	5.58	25.11
▶	Qtr - 4	5.45	24.52

Insights :-

- Sales Difference Between quarters is not Significant
- Q2 has been the highest Revenue Generator Quarter with the value (**5.61 CR**) ●
- Q2 and Q3 did not Perform As Expected ●
- Failed to enhance Sales in Summer holidays months and Festival season months Including Winter Holidays

```

with C1 as (
SELECT                                     -- adding metrics Quarterly for every year
    quarter_name,
    ROUND(SUM(CASE
        WHEN year = 2023 THEN total_amount_to_pay
    END) / 10000000,
    2) AS _2023_Revenue_in_CR,
    ROUND(SUM(CASE
        WHEN year = 2024 THEN total_amount_to_pay
    END) / 10000000,
    2) AS _2024_Revenue_in_CR
FROM
    fact_orders_enriched
GROUP BY quarter_name
),

c2 as (
SELECT                                     -- adding Vs PY_Values
    *,
    ROUND(((_2024_Revenue_in_CR - _2023_Revenue_in_CR) * 100 / _2023_Revenue_in_CR,
    2) AS vs_Prev_year_change
FROM
    c1
)
select * from c2;

```

Result 

	quarter_name	_2023_Revenue_in_CR	_2024_Revenue_in_CR	vs_Prev_year_change
►	Qtr - 1	2.80	2.78	-0.71
	Qtr - 2	2.81	2.80	-0.36
	Qtr - 3	2.76	2.82	2.17
	Qtr - 4	2.76	2.69	-2.54

Insights :-

- Sales Pattern is same in Both Years
- Q3 sales has icreased From last year by (+ 2.17 % | 6 Lakhs)
- Q1,Q2,Q3 sales has decreased on an Average by (- 1.23%) from the last year (YoY)
- Q4 has the highest Decrease in sales by (- 2.54 % | 7 Lakhs) from last year (YoY)
- Q3 and Q4 did not perform well (despite having Summer Holidays and festival Seasons)

4. Analyzing Monthly "Revenue & Demand" Trend (combined_for 2 years)

```

with c1 as (
SELECT                                     -- added Metics Monthly wise
    Month_name,
    ROUND(SUM(total_amount_to_pay) / 10000000, 2) AS Total_Revenue_CR,
    round(sum(quantity)/1000,2) as Demand_In_K,
    round(SUM(total_amount_to_pay)*100/(select sum(Total_amount_to_pay) from
fact_orders_enriched),2) as Contrirbution_to_overall_Rev
FROM
    fact_orders_enriched
GROUP BY month_name
),

c2 as (
select                                     -- added rolling_Rev_contribution
month_name,
Total_Revenue_CR,
Demand_In_K ,
sum(Contrirbution_to_overall_Rev) over( order by Contrirbution_to_overall_Rev desc) as
Rolling_Rev_contribution_PCT_Wise
from c1
)
select * from c2;

```

Result 

	month_name	Total_Revenue_CR	Demand_In_K	Rolling_Rev_contribution_PCT_Wise
▶	March	1.91	25.70	8.61
	January	1.91	25.68	17.20
	June	1.90	25.54	25.73
	May	1.89	25.67	34.23
	July	1.87	25.28	51.03
	October	1.87	25.23	51.03
	August	1.86	25.35	59.41
	September	1.85	24.80	67.75
	April	1.82	24.73	75.95
	December	1.81	24.32	84.09
	November	1.78	24.09	92.09
	February	1.76	23.74	100.01

Insights :-

- **Mar , Jan , Jun , May** are the Top 4 Performer months with an Average Revenue by (1.9 Cr) Covering (34 %) of Total Sales
- Sales Diff between the Best and bad performer Month is (15 Lakhs)
- **Nov and Feb** are the least performer months ●
- Summer Holidays and Festival months Did not Perform as Expected

5. Analyzing Monthly "Revenue & Demand" Trend YoY (For any Declination or Increment change)

with C1 as (

SELECT -- Added Rev & Dem Value monthly YoY

```
month_name,
ROUND(SUM(CASE
    WHEN year = 2023 THEN total_amount_to_pay
END) / 1000000,
2) AS _2023_Revenue_in_M,
ROUND(SUM(CASE
    WHEN year = 2024 THEN total_amount_to_pay
END) / 1000000,
2) AS _2024_Revenue_in_M,
ROUND(SUM(CASE
    WHEN year = 2023 THEN quantity
END) / 1000,
2) AS _2023_Demand_in_K,
ROUND(SUM(CASE
    WHEN year = 2024 THEN quantity
END) / 1000,
2) AS _2024_Demand_in_K
```

FROM

fact_orders_enriched

GROUP BY month_name

),

c2 as (

-- Added Vs PY Values

SELECT

*,

```
ROUND(( _2024_Revenue_in_M - _2023_Revenue_in_M ) * 100 / _2023_Revenue_in_M,
2) AS vs_Prev_yr_Rev_PCT_Wise,
ROUND(( _2024_Demand_in_K - _2023_Demand_in_K ) * 100 / _2023_Demand_in_K,
2) AS vs_Prev_year_Demand_PCT_Wise
```

FROM

c1

)

select * from c2;

Result 

	month_name	_2023_Revenue_in_M	_2024_Revenue_in_M	_2023_Demand_in_K	_2024_Demand_in_K	vs_Prev_yr_Rev_PCT_Wise	vs_Prev_year_Demand_PCT_Wise
▶	January	9.40	9.69	12.71	12.98	3.09	2.12
	February	8.74	8.85	11.85	11.89	1.26	0.34
	March	9.90	9.23	13.26	12.44	-6.77	-6.18
	April	9.23	8.99	12.59	12.14	-2.60	-3.57
	May	9.69	9.19	13.18	12.49	-5.16	-5.24
	June	9.14	9.81	12.22	13.32	7.33	9.00
	July	9.29	9.37	12.67	12.61	0.86	-0.47
	August	9.23	9.39	12.53	12.82	1.73	2.31
	September	9.05	9.48	12.16	12.64	4.75	3.95
	October	9.42	9.24	12.71	12.53	-1.91	-1.42
	November	9.00	8.78	12.22	11.87	-2.44	-2.86
	December	9.17	8.92	12.46	11.86	-2.73	-4.82

Insights :-

- Mar , Apr, May , Oct , Nov , Dec Month's Sales Decreased From last Year (YoY) 

- Mar , May sales dropped by Avg (- 5.9 %) YoY Worst Drop
- Apr , Oct , Nov , Dec Sales Avg Dropped by (- 2.43 %) from last Year (YoY)
- June Month Has the Highest Increase By (+ 7.33 %) YoY
- Month With Decrease in **Sales** also has a Decrease in **Demand** with almost Same percentage as Rev Except **Dec** (Demand Dropped (2 times) the the Rev PCT Wise)
- June month has the Highest Demand Increase By (+ 9 %) YoY

6. objective : Analyzing Weekly Trend (combined for 2 years) (Customer_preference)

With C1 as (

```
SELECT -- added needed metrics Weekly
    week_of_month,
    ROUND(SUM(total_amount_to_pay) / 10000000, 2) AS Total_Revenue_CR,
    round(SUM(total_amount_to_pay)*100 /( select sum(total_amount_to_pay) from
fact_orders_enriched),2) as PCT_contribution_overall_Sales
FROM
    fact_orders_enriched
GROUP BY week_of_month
),
```

c2 as (

```
-- Adding rolling_Rev_Contribution_PCT_wise
select week_of_month,
Total_Revenue_CR,
sum(PCT_contribution_overall_Sales) over(order by PCT_contribution_overall_Sales desc ) as
Rolling_Rev_contribution_PCT_wise
from c1
)
select * from c2;
```

Result 

	week_of_month	Total_Revenue_CR	Rolling_Rev_contribution_PCT_wise
▶	2	5.31	23.88
	4	5.05	46.62
	3	4.97	68.98
	5	3.72	85.72
	1	2.98	99.14
	6	0.19	100.00

Insights :-

- Week 2 is the Highest Revenue Generator with the Value (5.31 Cr)
- Week 2 ,3, 4 are Covering Almost **70%** of Total Revenue (Customer Prefer these weeks)
- Week 6 is the Week in which People Prefer not to shop More only Generated (0.19 Cr)

C . WeekDays to Hourly Trend Analysis

7. **objective : Analyzing WeekDay's "Revenue & Demand" Trend (Customer_Preference) combine 2 years for overall View**

```
with c1 as(
SELECT                                     -- added needed metrics Weekdays wise
    Day_name,
    Day_of_week,
    ROUND(SUM(total_amount_to_pay) / 10000000, 2) AS Total_Revenue_CR,
    ROUND(SUM(quantity) / 1000, 2) AS Demand_In_K,
    round(SUM(total_amount_to_pay)*100 /( select sum(total_amount_to_pay) from fact_orders_enriched),2) as
PCT_contribution_overall_Sales

FROM
    fact_orders_enriched
GROUP BY Day_name , Day_of_week
ORDER BY Day_of_week
),

c2 as (                                     -- added Rolling Rev Contribution PCT wise
    select
        Day_name,
        Day_of_week,
        Total_Revenue_CR,
        Demand_In_K,
        sum(PCT_contribution_overall_Sales) over(order by PCT_contribution_overall_Sales desc ) as
Rolling_Rev_contribution_PCT_wise
    from c1
)
select * from c2;
```

Result

Day_name	Day_of_week	Total_Revenue_CR	Demand_In_K	Rolling_Rev_contribution_PCT_wise
Wednesday	3	3.21	43.28	14.45
Sunday	7	3.21	43.08	28.88
Saturday	6	3.19	42.97	43.24
Friday	5	3.18	42.84	57.54
Monday	1	3.16	42.93	86.02
Tuesday	2	3.16	42.84	86.02
Thursday	4	3.10	42.18	99.99

Insights :-

- **Wed** , and **Weekends** are the top performer in terms of Revenue & Demand Generator Days 
- As weekends Comes Customers shops more 
- Rest days have descent Performance 
- Top 3 Rev Generator Days Covers (**43 %**) of Total Revenue

8. Objective : Analyzing WeekEnd vs WeekDay "Revenue & Demand & AOV(Average order Value)" Trend

```
with c1 as (                                -- added needed metrics weekend and days wise
SELECT
    Week_End_or_day,
    ROUND(SUM(total_amount_to_pay) / 10000000, 2) AS Total_Revenue_CR,
    ROUND(SUM(quantity) / 1000, 2) AS Demand_In_K,
    ROUND(COUNT(DISTINCT (order_id)) / 1000, 2) AS Total_Uniq_Ordr_in_K
FROM
    fact_orders_enriched
GROUP BY Week_End_or_day
),

c2 as (                                     -- Added Rev grouped by order id and order_day
SELECT
    order_id,
    Week_End_or_day,
    SUM(total_amount_to_pay) AS total_Amount_to_Pay
FROM
    fact_orders_enriched
GROUP BY order_id , Week_End_or_day
)
,
c3 as (                                     -- added AOV grouped by weekend/day
SELECT
    Week_End_or_day,
    ROUND(AVG(total_Amount_to_Pay) / 1000, 2) AS AOV
FROM c2
GROUP BY Week_End_or_day
)
SELECT * FROM c1
JOIN
    c3 USING (Week_End_or_day);
```

Result 

Week_End_or_day	Total_Revenue_CR	Demand_In_K	Total_Uniq_Ordr_in_K	AOV
Weekday	19.02	257.04	28.56	6.66
Weekend	3.21	43.08	4.81	6.67

Insights :-

- Weekend and WeekDays behaviour is same as Expected
- Weekend Performance is not same as Expected or it should be
- AOV is same for weekend and Weekdays (6.66 K)

9. Objective : Analyzing "Demand" Trend using Hourly_Buckets

```
with c1 as (  
    SELECT                                     -- added Demand value Hourly  
    HOUR(order_time),  
    CASE  
        WHEN HOUR(order_time) BETWEEN 1 AND 3 THEN '1 - 3'  
        WHEN HOUR(order_time) BETWEEN 4 AND 6 THEN '4 -6'  
        WHEN HOUR(order_time) BETWEEN 7 AND 9 THEN '7 to 9'  
        WHEN HOUR(order_time) BETWEEN 10 AND 12 THEN '10 - 12'  
        WHEN HOUR(order_time) BETWEEN 13 AND 15 THEN '13 - 15'  
        WHEN HOUR(order_time) BETWEEN 16 AND 18 THEN '16- 18'  
        WHEN HOUR(order_time) BETWEEN 19 AND 21 THEN '17 - 21'  
        ELSE '22 -24'  
    END AS Hourly_Bucket,  
    quantity  
FROM  
    fact_orders_enriched  
)  
SELECT                                     -- Result  
    Hourly_Bucket,  
    ROUND(SUM(quantity) / 1000, 2) AS Demand_In_K  
FROM  
    c1  
GROUP BY Hourly_Bucket  
ORDER BY Demand_In_K DESC;
```

Result 

	Hourly_Bucket	Demand_In_K
▶	16- 18	38.48
	22 -24	38.26
	7 to 9	37.97
	1 - 3	37.59
	10 - 12	37.13
	17 - 21	37.02
	4 -6	37.01
	13 - 15	36.66

Insights :-

- Demand Difference is not that Significant between Diff Time slots 
- The Demand Difference highest demand and lowest demand is only (1.82 k)
- 16 -18 slot is the slot with the highest Demand with the value (38.48 K)
- Morning (7 - 9) , Evening (16 -18) and Night time (22 -24) slot are Top 3 slots 
- Rest Time slots are also medium performer

Product Level Insights and Recommendation

Product Level Insights and Recommendations -

KPI 's : -

1.#_Unique_Products

```
SELECT  
  COUNT(DISTINCT (product_id)) AS Total_Unique_Products_sold_yet  
FROM  
  fact_orders_enriched;
```

	Total_Unique_Products_
▶	237

2. #_Revenue_Generated

```
select round(sum( total_amount_to_pay)/10000000,2) as Revenue_CR from  
fact_orders_enriched;
```

	Total_Rev_IN_CR
▶	22.22

3. #_PROFIT_Generated

```
select Round((sum( total_amount_to_pay) - sum(total_Cost_price))/10000000,2) as Profit_CR  
from fact_orders_enriched;
```

	Profit_CR
▶	6.70

4. #_PROFIT_Percentage

```
select Round((sum( total_amount_to_pay) -  
sum(total_Cost_price))/(sum(total_amount_to_Pay)),2)*100 as Profit_PCT from  
fact_orders_enriched;
```

Categorical Level Insights : -

1 . objective : Analyzing Categorical Level Total "Revenue & Demand & Profit "

```

with c1 as (
SELECT
    category AS Product_Category,
    ROUND(SUM(quantity) / 1000, 2) AS Total_QTY_sold_in_K,
    ROUND(SUM(total_amount_to_pay) / 10000000, 2) AS Revenue__IN_CR,
    (ROUND(SUM(total_amount_to_pay) / 10000000, 2) - ROUND(SUM(total_cost_price) / 10000000, 2)) AS Total_Profit_In_CR
FROM
    fact_orders_enriched
GROUP BY Product_Category
),
    c2 as (
SELECT
    *,
    ROUND((Total_Profit_In_CR * 100) / Revenue__IN_CR,
    2) AS Profit_Margin,

    ROUND((Revenue__IN_CR * 100) / (select sum(total_Amount_to_pay)/10000000 from fact_orders_enriched),
    2) As "%_contribution_overall_Sales"

FROM
    c1
)
SELECT * FROM c2;

```

Result

	Product_Category	Total_QTY_sold_in_K	Revenue__IN_CR	Total_Profit_In_CR	Profit_Margin	%_contribution_overall_Sales
▶	Instant & Frozen Food	37.53	3.48	1.39	39.94	15.66
	Baby Care	38.15	2.88	0.86	29.86	12.96
	Cold Drinks & Juices	33.32	2.81	0.83	29.54	12.64
	Pet Care	30.50	2.59	0.89	34.36	11.65
	Snacks & Munchies	24.08	2.04	0.72	35.29	9.18
	Fruits & Vegetables	30.29	1.98	0.49	24.75	8.91
	Household Care	22.90	1.61	0.40	24.84	7.24
	Personal Care	18.99	1.46	0.51	34.93	6.57
	Grocery & Staples	25.46	1.27	0.18	14.17	5.71
	Dairy & Breakfast	20.12	1.13	0.22	19.47	5.08
	Pharmacy	18.78	0.96	0.19	19.79	4.32

Insights : -

- **Instant & frozen food** is the Top Performing Category (Rev - 3.48 Cr , Profit - 1.39 Cr , Qty_sold - 37.53 K) 
- Top 5 Category Generates Almost (62 %) of Total Revenue
- **Pharmacy** is least Revenue generator with the value (0.19 cr) only with the contribution percentage (4.32 %) to the overall revenue
- Bottom 3 categories are only ones with profit_margin < 20 % and also the Rev_contribution <= 5.5 % 
- Demand Wise Every category has been Medium to High demand
- **Instant & frozen food** is the one which is Top in terms of (Rev , Demand , Profit , Profit_Margin)

2 - objective : Analyzing Product_Category 's "Revenue & Demand & Profit" Performance "YoY"

with c1 as (

SELECT

category AS Product_Category,

ROUND(SUM(CASE

WHEN year = 2023 THEN quantity

END) / 1000,

2) AS QTY_sold_in_K_in_2023,

ROUND(SUM(CASE

WHEN year = 2023 THEN total_amount_to_pay

END) / 10000000,

2) Revenue_IN_CR_in_2023,

(ROUND(SUM(CASE

WHEN year = 2023 THEN total_amount_to_pay

END) / 10000000,

2) - ROUND(SUM(CASE

WHEN year = 2023 THEN total_cost_price

END) / 10000000,

2)) AS _Profit_In_CR_in_2023,

ROUND(SUM(CASE

WHEN year = 2024 THEN quantity

END) / 1000,

2) AS QTY_sold_in_K_in_2024,

ROUND(SUM(CASE

WHEN year = 2024 THEN total_amount_to_pay

END) / 10000000,

2) Revenue_IN_CR_in_2024,

(ROUND(SUM(CASE

WHEN year = 2024 THEN total_amount_to_pay

END) / 10000000,

2) - ROUND(SUM(CASE

WHEN year = 2024 THEN total_cost_price

END) / 10000000,

2)) AS _Profit_In_CR_in_2024

FROM

fact_orders_enriched

GROUP BY Product_Category

),

c2 as (

SELECT

*,

ROUND((_Profit_In_CR_in_2023 * 100) / Revenue_IN_CR_in_2023,

2) AS Profit_Margin_in_2023,

ROUND((_Profit_In_CR_in_2024 * 100) / Revenue_IN_CR_in_2024,

2) AS Profit_Margin_in_2024

from c1),

c3 as (

SELECT

*,

ROUND((QTY_sold_in_K_in_2024 - QTY_sold_in_K_in_2023) * 100 / QTY_sold_in_K_in_2023,

2) AS vs_Previous_year_Demand,

ROUND((Revenue_IN_CR_in_2024 - Revenue_IN_CR_in_2023) * 100 / Revenue_IN_CR_in_2023,

2) AS vs_Previous_year_Revenue,

ROUND((Profit_Margin_in_2024 - Profit_Margin_in_2023) * 100 / Profit_Margin_in_2023,

2) AS vs_Previous_year_profit_margin

FROM

```

c2
)
SELECT
Product_Category,
vs_Previous_year_Demand,
vs_Previous_year_Revenue,
vs_Previous_year_profit_margin
FROM
c3;

```

Result 

Product_Category	vs_Previous_year_Demand	vs_Previous_year_Revenue	vs_Previous_year_profit_margin
Instant & Frozen Food	-0.27	-0.57	0.58
Personal Care	-2.39	0.00	0.00
Baby Care	0.16	0.00	0.00
Fruits & Vegetables	-3.06	-3.00	-1.04
Snacks & Munchies	1.00	1.98	0.87
Pharmacy	-2.52	0.00	11.09
Pet Care	-1.63	0.78	1.50
Dairy & Breakfast	4.68	5.45	13.81
Cold Drinks & Juices	-1.67	-2.11	-0.27
Grocery & Staples	-1.56	-1.56	1.64
Household Care	1.23	2.50	2.44

Insights :-

1. Revenue -

- “Dairy & BreakFast” with the Highest Increase in Rev by (+ 5.45 %) from last year
- Category 2,3 6 are with the lowest increase with (+ 0.00 %) from Last Year and rest were between it
- 4 categories’s Revenue dropped from Last Year out of 11 
- “Fruits and Vegetables” with the highest drop from 4 with (- 3 %)
- “Instant & frozen food” is the only one with lowest drop from 4 by (- 0.57 %)
- Rest 2 are dropped by >= 1 to < 3 %

2. Demand -

- “Dairy & breakfast” is with the highest increase with (+ 4.68 %) from last year
- “Baby care” with the least increase with (+ 0.16 %) and rest were between it
- 7 Categories ‘s Demand dropped from last year out of 11 
- “Fruits and Vegetables” with the highest drop from 7 with (- 3.06 %)
- “Instant & frozen food” is the only one with lowest drop from 7 by (- 0.27 %)
- Rest 5 are dropped by >=1 and < 3 % from last year

3. Profit margin

- “Dairy & breakfast” and “Pharmacy” is with the Highest Profit_Margin Increase with (+ 13.81 % and + 11.09 % of Rev_Gen) From Last year
- “Personal care” and “baby care” with the least increase with (+ 0.00%) and rest increase were in between them
- 2 categories ‘s Profit margin went down from last year 
- “Fruits and Vegetables” with highest decrease with (-1.04 %)
- “Cold drinks & juices” with least decrease with (-0.27 %) from last year

3. -- Objective : Analyzing the product_Categories Monthly Trend To See Seasoanal_pattern if Any

```
SELECT
    category AS Product_Category,
    month,
    ROUND(SUM(quantity) / 1000, 2) AS Total_Demand_in_K,
    ROUND(SUM(total_amount_to_pay) / 1000000, 2) AS Revenue__in_M
FROM
    fact_orders_enriched
GROUP BY Product_Category , month_
ORDER BY Product_Category , month_;
```

Result

Month	Baby Care	Cold Drinks & Juices	Dairy & Breakfast	Fruits & Vegetables	Grocery & Staples	Household Care	Instant & Frozen Food	Personal Care	Pet Care	Pharmacy	Snacks & Munchies
1	₹ 2.5 M	₹ 2.4 M	₹ 0.9 M	₹ 1.6 M	₹ 1.1 M	₹ 1.4 M	₹ 3.1 M	₹ 1.2 M	₹ 2.2 M	₹ 0.8 M	₹ 1.8 M
2	₹ 2.3 M	₹ 2.2 M	₹ 0.9 M	₹ 1.6 M	₹ 1.0 M	₹ 1.2 M	₹ 2.7 M	₹ 1.2 M	₹ 2.1 M	₹ 0.8 M	₹ 1.6 M
3	₹ 2.6 M	₹ 2.4 M	₹ 1.0 M	₹ 1.6 M	₹ 1.1 M	₹ 1.5 M	₹ 3.0 M	₹ 1.3 M	₹ 2.1 M	₹ 0.8 M	₹ 1.8 M
4	₹ 2.3 M	₹ 2.2 M	₹ 0.9 M	₹ 1.7 M	₹ 1.1 M	₹ 1.3 M	₹ 2.9 M	₹ 1.3 M	₹ 2.2 M	₹ 0.8 M	₹ 1.6 M
5	₹ 2.5 M	₹ 2.4 M	₹ 1.0 M	₹ 1.6 M	₹ 1.1 M	₹ 1.4 M	₹ 3.0 M	₹ 1.2 M	₹ 2.2 M	₹ 0.8 M	₹ 1.7 M
6	₹ 2.4 M	₹ 2.5 M	₹ 1.0 M	₹ 1.7 M	₹ 1.1 M	₹ 1.4 M	₹ 2.8 M	₹ 1.2 M	₹ 2.2 M	₹ 0.8 M	₹ 1.8 M
7	₹ 2.4 M	₹ 2.4 M	₹ 0.9 M	₹ 1.7 M	₹ 1.1 M	₹ 1.3 M	₹ 3.1 M	₹ 1.3 M	₹ 2.0 M	₹ 0.8 M	₹ 1.8 M
8	₹ 2.3 M	₹ 2.3 M	₹ 1.0 M	₹ 1.7 M	₹ 1.1 M	₹ 1.4 M	₹ 2.9 M	₹ 1.2 M	₹ 2.3 M	₹ 0.8 M	₹ 1.7 M
9	₹ 2.4 M	₹ 2.4 M	₹ 0.9 M	₹ 1.6 M	₹ 1.0 M	₹ 1.3 M	₹ 2.9 M	₹ 1.3 M	₹ 2.2 M	₹ 0.8 M	₹ 1.7 M
10	₹ 2.4 M	₹ 2.5 M	₹ 1.0 M	₹ 1.7 M	₹ 1.1 M	₹ 1.4 M	₹ 2.9 M	₹ 1.2 M	₹ 2.2 M	₹ 0.8 M	₹ 1.6 M
11	₹ 2.3 M	₹ 2.3 M	₹ 0.9 M	₹ 1.6 M	₹ 1.0 M	₹ 1.3 M	₹ 2.8 M	₹ 1.1 M	₹ 2.1 M	₹ 0.8 M	₹ 1.6 M
12	₹ 2.4 M	₹ 2.3 M	₹ 0.9 M	₹ 1.6 M	₹ 1.0 M	₹ 1.3 M	₹ 2.9 M	₹ 1.1 M	₹ 2.2 M	₹ 0.8 M	₹ 1.7 M

Insights : -

- 0 out 11 categories can be seen following any seasonal pattern
- Just some categories generating more monthly revenue then other individual category
- Every Category is generating almost same amount of Rev every month

Product Level Insights :-

1. objective : finding Top 20 Performing Products in terms of "Revenue" with their contribution to overall Revenue with Rolling

```
with c1 as (
SELECT
    Product_name,
    category AS product_Category,
    ROUND(SUM(total_amount_to_pay) / 100000, 2) AS Revenue__in_Lakhs
FROM
    fact_orders_enriched
GROUP BY Product_name , product_Category
ORDER BY Revenue__in_Lakhs DESC
LIMIT 20
),
c2 as (
select *,
round((Revenue__in_Lakhs*100)/(select sum(total_amount_to_pay)/100000 from fact_orders_enriched),2) as
Contribution_to__overall
from c1
),
c3 as (
    select product_name,product_Category,Revenue__in_Lakhs ,
    sum(Contribution_to__overall) over (order by Contribution_to__overall desc) as
Rolling_contribution_to_overall_Rev
from c2
)
select * from c3;
```

Inshighs :

- Top 20 products covers (**16 %**) of Total_Revenue
- In Top 20 - instant & frozen Food ,Snacks & Munchies and Baby care Covers (**75 %**) of Products
- In top 20, 6 product is from instant & frozen food category as highest and bottom 3 with 1 as least

Metrix View 

product_Category	No_of_Products
Instant & Frozen Food	6
Snacks & Munchies	5
Baby Care	4
Personal Care	2
Pet Care	1
Household Care	1
Cold Drinks & Juices	1

product_name	product_Category	Revenue__in_Lakhs	Rolling_contribution_to_overall_Rev
iure repudiandae	Instant & Frozen Food	19.87	0.89
facere nesciunt	Snacks & Munchies	18.78	1.74
ea sequi	Baby Care	18.75	3.42
consequuntur ab	Instant & Frozen Food	18.70	3.42
inventore ab	Snacks & Munchies	18.45	5.08
labore magnam	Instant & Frozen Food	18.36	5.08
perferendis magnam	Personal Care	18.13	5.90
est quos	Snacks & Munchies	18.00	6.71
exercitationem provident	Instant & Frozen Food	17.87	9.11
consectetur magnam	Baby Care	17.79	9.11
voluptates quam	Snacks & Munchies	17.73	9.11
corporis repudiandae	Baby Care	17.47	9.90
neque aut	Snacks & Munchies	17.43	11.46
omnis ipsam	Instant & Frozen Food	17.29	11.46
consequatur qui	Instant & Frozen Food	17.21	15.31
ipsam distinctio	Pet Care	17.20	15.31
aperiam explicabo	Household Care	17.11	15.31
nostrum tempore	Cold Drinks & Juices	17.11	15.31
cum blanditiis	Personal Care	17.05	15.31
error sapiente	Baby Care	16.90	16.07

2. Objective : finding Bottom 20 Products in terms of "Revenue Generating"

```

SELECT
product_name,
category AS product_Category,
ROUND(SUM(total_Amount_to_pay) / 100000, 2) AS Revenue_in_lakhs
FROM
fact_orders_enriched
GROUP BY product_name , product_Category
ORDER BY Revenue_in_lakhs Asc limit 20;

```

product_name	product_Category	Revenue_in_lakhs
corporis nesciunt	Pharmacy	0.36
repellendus tempora	Snacks & Munchies	0.40
laborum saepe	Grocery & Staples	0.48
doloribus laudantium	Household Care	0.54
dolore ea	Fruits & Vegetables	0.55
corrupti atque	Fruits & Vegetables	0.58
numquam corporis	Grocery & Staples	0.60
qui in	Personal Care	0.60
provident dolorum	Instant & Frozen Food	0.79
reprehenderit ea	Instant & Frozen Food	0.84
dicta vero	Grocery & Staples	0.86
occaecati eaque	Personal Care	1.06
optio minima	Dairy & Breakfast	1.11
ab sed	Pharmacy	1.23
optio sunt	Pharmacy	1.30
necessitatibus rem	Fruits & Vegetables	1.46
necessitatibus temporibus	Household Care	1.71
quis quae	Baby Care	1.72
distinctio tenetur	Pet Care	1.74
dolorem corrupti	Fruits & Vegetables	1.76

Insights : -

- Bottom 20 covers only (0.89 %) of total_Rev
- In this , Pharmacy , Fruits & vegetable And Grocery & staples Category (3 out of 10) covers (50%) of bottom 20 products
- highest 4 products of Fruits and Vegetables category and And bottom 4 with 1 products as shown
- Coporiss nesclunt from the pharmacy Category has been the least Revenue generating Product with 0.36 lakhs /36 K

product_Category	No_of_Prouducts
Fruits & Vegetables	4
Pharmacy	3
Grocery & Staples	3
Household Care	2
Personal Care	2
Instant & Frozen Food	2
Snacks & Munchies	1
Dairy & Breakfast	1
Baby Care	1
Pet Care	1

3. - Objective : finding Top 20 Products in terms of " Profit Making"

```
SELECT
    Product_name,
    category AS product_Category,
    ROUND(
SUM(total_amount_to_pay - total_cost_price ) /
100000, 2) AS Profit_in_Lakhs
FROM
    fact_orders_enriched
GROUP BY Product_name , product_Category
ORDER BY Profit_in_Lakhs DESC
LIMIT 20
```

Product_name	product_Category	Profit_in_Lakhs
iure repudiandae	Instant & Frozen Food	8.03
consequuntur ab	Instant & Frozen Food	7.47
exercitationem provident	Instant & Frozen Food	7.22
labore magnam	Instant & Frozen Food	7.00
omnis ipsam	Instant & Frozen Food	6.92
nemo libero	Instant & Frozen Food	6.65
consequatur qui	Instant & Frozen Food	6.62
facere nesciunt	Snacks & Munchies	6.60
perferendis magnam	Personal Care	6.59
neque aut	Snacks & Munchies	6.57
aperiam aspernatur	Instant & Frozen Food	6.49
facilis vitae	Instant & Frozen Food	6.34
cum blanditiis	Personal Care	6.12
est quos	Snacks & Munchies	6.10
inventore ab	Snacks & Munchies	6.08
voluptates quam	Snacks & Munchies	5.97
ipsam distinctio	Pet Care	5.85
harum aut	Snacks & Munchies	5.79
eos laudantium	Instant & Frozen Food	5.78
enim odit	Personal Care	5.78

3.5 Objective : finding Bottom 20 Products in terms of "Profit making"

```
SELECT
    product_name,
    category AS product_Category,
    ROUND((SUM(total_amount_to_pay) -
SUM(total_cost_price))/100000,
2) AS profit_in_Lakhs
FROM
    fact_orders_enriched
GROUP BY product_name , product_Category
ORDER BY profit_in_Lakhs Asc limit 20;
```

product_name	product_Category	profit_in_Lakhs
laborum saepe	Grocery & Staples	0.06
corporis nesciunt	Pharmacy	0.07
numquam corporis	Grocery & Staples	0.09
dicta vero	Grocery & Staples	0.12
corrupti atque	Fruits & Vegetables	0.13
doloribus laudantium	Household Care	0.13
dolore ea	Fruits & Vegetables	0.14
repellendus tempora	Snacks & Munchies	0.14
laudantium autem	Grocery & Staples	0.18
qui in	Personal Care	0.20
optio minima	Dairy & Breakfast	0.20
ab sed	Pharmacy	0.24
optio sunt	Pharmacy	0.26
placeat eveniet	Grocery & Staples	0.27
provident dolorum	Instant & Frozen ...	0.32
praesentium totam	Dairy & Breakfast	0.33
reprehenderit ea	Instant & Frozen ...	0.34
necessitatibus rem	Fruits & Vegetables	0.34
reprehenderit molestias	Grocery & Staples	0.35
occaecati eaque	Personal Care	0.37

Top 20's - Insights :-

1. This top 20 Products covers (19.40 %) of Total Profit Generated
2. Instant & Frozen food and Snacks & Munchies covers (80 %) of Products of top 20 And rest 20 %
3. Lure repudaindae from Instant & Frozen food Category with the highest profit with (8.03 Lakhs)

product_Category	no_of_Products
Instant & Frozen Food	10
Snacks & Munchies	6
Personal Care	3
Pet Care	1

Bottom 20's insights :-

1. This Bottom 20 covers only (0.64 %) of total Profit
2. Grocery & staples , Pharmacy and Fruit & vegetables covers (60 %) of Bottom 20 Products
3. Laburum saepe from grocery & staple category is the product with the least Profit (0.06 lakhs/ 6 K)

product_Category	No_of_Products
Grocery & Staples	6
Pharmacy	3
Fruits & Vegetables	3
Personal Care	2
Dairy & Breakfast	2
Instant & Frozen Food	2
Household Care	1
Snacks & Munchies	1

4. -- Objective : finding Top 20 highest selling products (QTY sold wise)

```
SELECT
    product_name,
    category AS product_Category,
    ROUND(SUM(quantity) / 1000, 2) AS
Total_Units_sold_K
FROM
    fact_orders_enriched
GROUP BY product_name ,
product_Category
ORDER BY Total_Units_sold DESC limit 20;
```

product_name	product_Category	Total_Units_sold
ipsum impedit	Cold Drinks & Juices	1.42
voluptas earum	Grocery & Staples	1.41
quasi corrupti	Fruits & Vegetables	1.41
sapiente facilis	Pet Care	1.40
provident sapiente	Baby Care	1.40
perferendis magnam	Personal Care	1.38
quibusdam in	Baby Care	1.38
voluptates repellendus	Baby Care	1.38
totam reprehenderit	Pet Care	1.38
dolore ea	Fruits & Vegetables	1.37
impedit dicta	Personal Care	1.37
maiores esse	Fruits & Vegetables	1.36
maiores eveniet	Cold Drinks & Juices	1.36
corporis repudiandae	Baby Care	1.36
architecto ipsam	Cold Drinks & Juices	1.35
ut dolorum	Grocery & Staples	1.35
dicta vero	Grocery & Staples	1.35
voluptates quam	Snacks & Munchies	1.34
eaque cupiditate	Baby Care	1.34
soluta enim	Pet Care	1.33

4.5 finding Bottom 20 Products in terms of Demand / Qty sold

```
SELECT
    product_name,
    category AS product_Category,
    ROUND(SUM(quantity) / 1000, 2) AS
Total_Units_sold
FROM
    fact_orders_enriched
GROUP BY product_name , product_Category
ORDER BY Total_Units_sold Asc limit 20;
```

product_name	product_Category	Total_Units_sold_k
placeat eveniet	Grocery & Staples	1.11
molestiae iure	Household Care	1.16
soluta facere	Dairy & Breakfast	1.16
repellat molestias	Pharmacy	1.16
fugiat magnam	Fruits & Vegetables	1.17
quibusdam quas	Instant & Frozen Food	1.17
velit sit	Cold Drinks & Juices	1.17
blanditiis non	Baby Care	1.18
distinctio natus	Cold Drinks & Juices	1.18
ut cum	Dairy & Breakfast	1.18
aliquam expedita	Dairy & Breakfast	1.18
ea aliquam	Instant & Frozen Food	1.18
dignissimos exercitationem	Pet Care	1.18
libero earum	Baby Care	1.19
doloribus laudantium	Household Care	1.19
nemo delectus	Grocery & Staples	1.19
optio vitae	Baby Care	1.19
architecto sapiente	Pet Care	1.19
facilis vitae	Instant & Frozen Food	1.19
placeat provident	Fruits & Vegetables	1.19

Top 20's insights :-

1. Ipsum impedit from cold drinks & juices category is the product got sold highest in QTY With (1.42 k)
2. In Top 20 , (55 %) of Products are covered by top 3 category shown in fgr below
3. Only 1 product of Snacks & munchies Category

product_Category	no_of_Products
Baby Care	5
Cold Drinks & Juices	3
Grocery & Staples	3
Fruits & Vegetables	3
Pet Care	3
Personal Care	2
Snacks & Munchies	1

Bottom 20's insight

1. Placet Eveniet is the product from Grocery & staples Category with the Least Qty Sold (1.11 K)
2. In bottom 20 , only 1 product from Pharmacy category
3. Top 3 from below fgr covers (45 %)

product_Category	no_of_Products
Dairy & Breakfast	3
Instant & Frozen Food	3
Baby Care	3
Grocery & Staples	2
Household Care	2
Fruits & Vegetables	2
Cold Drinks & Juices	2
Pet Care	2
Pharmacy	1

5. objective : Top 20 Products with "High sales"  with "High_Profit_Margin" 

```
SELECT
product_name,
category AS product_Category,
ROUND(SUM(total_amount_to_pay) / 100000, 2) AS Revenue_in_lakhs,
ROUND((SUM(total_amount_to_pay) - SUM(total_cost_price)) * 100 /
SUM(total_amount_to_pay),
2) AS profit_margin
FROM
fact_orders_enriched
GROUP BY product_name , product_Category
HAVING Revenue_in_lakhs >= 8
AND profit_margin >= 25
ORDER BY Revenue_in_lakhs DESC;
```

Result 

product_name	product_Category	Revenue_in_lakhs	profit_margin
iure repudiandae	Instant & Frozen Food	19.87	40.43
facere nesciunt	Snacks & Munchies	18.78	35.16
ea sequi	Baby Care	18.75	30.75
consequuntur ab	Instant & Frozen Food	18.70	39.94
inventore ab	Snacks & Munchies	18.45	32.95
labore magnam	Instant & Frozen Food	18.36	38.10
perferendis magnam	Personal Care	18.13	36.35
est quos	Snacks & Munchies	18.00	33.91
exercitationem provident	Instant & Frozen Food	17.87	40.41
consectetur magnam	Baby Care	17.79	28.12
voluptates quam	Snacks & Munchies	17.73	33.70
corporis repudiandae	Baby Care	17.47	29.84
neque aut	Snacks & Munchies	17.43	37.70
omnis ipsam	Instant & Frozen Food	17.29	40.04
consequatur qui	Instant & Frozen Food	17.21	38.48
ipsam distinctio	Pet Care	17.20	34.03
nostrum tempore	Cold Drinks & Juices	17.11	28.68
aperiam explicabo	Household Care	17.11	25.68
cum blanditiis	Personal Care	17.05	35.90
error sapiente	Baby Care	16.90	30.06

"HighSales_with_HighProfit_full_ProductList :-

https://drive.google.com/file/d/1G_50KhE6_ApwXeSVGDmTLkTYHzjmr2K/view?usp=sharing

Insights : -

- product with "high_Sales_with_highMargin" covers (16.08 %) of Total_Revenue
- In this "instant & Frozen food", "Snacks & munches" and "Baby Care" Category Covers (75 %) of Products
- These Products having an Average Profit Margin Of (34.51 %) of Revenue Generated

product_Category	No_of_Product
Instant & Frozen Food	6
Snacks & Munchies	5
Baby Care	4
Personal Care	2
Pet Care	1
Cold Drinks & Juices	1
Household Care	1

6. objective : Top 20 products with the "High Sales" but "Low Profit_Margin "

```
SELECT
    product_name,
    category AS product_Category,
    ROUND(SUM(total_amount_to_pay) / 100000, 2) AS Revenue_in_lakhs,
    ROUND((SUM(total_amount_to_pay) - SUM(total_cost_price)) * 100 /
SUM(total_amount_to_pay),
        2) AS profit_margin
FROM
    fact_orders_enriched
GROUP BY product_name , product_Category
HAVING Revenue_in_lakhs >= 8
    AND profit_margin <=15
ORDER BY Revenue_in_lakhs DESC;
```

Result 

	product_name	product_Category	Revenue_in_lakhs	profit_margin
▶	recusandae ad	Grocery & Staples	14.13	14.07
	voluptas earum	Grocery & Staples	13.47	13.16
	dicta maxime	Grocery & Staples	11.81	14.44
	ut dolorum	Grocery & Staples	8.98	12.39

Insights : -

- These four products Covering (2.18 %) of total_Rev
- Having an Average Profit_Margin Of only (13.52 %)
- 100% Products are from Grocery & Stables Category
- Ut dolorum Product tops in product_with_highsales (8.98 Lakshs) with lowest profit_margin of (12.39 %)

7. objective : Finding Top 20 products with the "Low Sales" but "High Profit_Margin "

```
SELECT
    product_name,
    category AS product_Category,
    ROUND(SUM(total_amount_to_pay) / 100000, 2) AS Revenue_in_lakhs,
    ROUND((SUM(total_amount_to_pay) - SUM(total_cost_price)) * 100 / SUM(total_amount_to_pay),
        2) AS profit_margin
FROM
    fact_orders_enriched
GROUP BY product_name , product_Category
HAVING Revenue_in_lakhs <=7
    AND profit_margin >=25
ORDER BY Revenue_in_lakhs Asc;
```

Result 

product_name	product_Category	Revenue_in_lakhs	profit_margin
repellendus tempora	Snacks & Munchies	0.40	35.07
dolore ea	Fruits & Vegetables	0.55	26.41
qui in	Personal Care	0.60	32.51
provident dolorum	Instant & Frozen Food	0.79	40.59
reprehenderit ea	Instant & Frozen Food	0.84	40.52
occaecati eaque	Personal Care	1.06	34.90
quis quae	Baby Care	1.72	29.79
distinctio tenetur	Pet Care	1.74	33.90
maiores eveniet	Cold Drinks & Juices	1.81	28.46
laboriosam porro	Snacks & Munchies	1.99	35.07
sunt amet	Instant & Frozen Food	2.20	39.81
eveniet fugiat	Baby Care	2.23	28.52
quod accusamus	Baby Care	2.27	29.08
suscipit repellendus	Personal Care	2.47	31.56
vitae ipsa	Cold Drinks & Juices	2.55	31.24
sit ipsam	Instant & Frozen Food	2.56	37.97
soluta officia	Cold Drinks & Juices	2.72	27.97
quis optio	Snacks & Munchies	2.80	34.18
doloremque deleniti	Baby Care	3.22	29.42
ipsum impedit	Cold Drinks & Juices	3.36	31.96

"Lowsales_with_HighSales_ProductList"

https://drive.google.com/file/d/1mGQP0Jr6imPMc2jw56EiGys6KmDd2Flw/view?usp=drive_link

Insights : -

- These Products Covering only (2.43 %) of Total_Rev
- Having an Average Profit_Margin of (32.20 %) of Revenue Generated
- "Instant & Frozen Food" , "Baby Care" and "cold drinks & juices " category covering (60 %) of Above 20 products
- Provident Dolorum from "Instant & Frozen Food" Category toposes the product in lowSales (0.79 Lakh) with highMargin (40.59 %)

product_Category	No_of_Products
Instant & Frozen Food	4
Baby Care	4
Cold Drinks & Juices	4
Snacks & Munchies	3
Personal Care	3
Fruits & Vegetables	1
Pet Care	1

8. objective : Finding the Top 20 products with the "Low Sales" and "Low Profit_Margin "

```
SELECT
    product_name,
    category AS product_Category,
    ROUND(SUM(total_amount_to_pay) / 100000, 2) AS Revenue_in_lakhs,
    ROUND((SUM(total_amount_to_pay) - SUM(total_cost_price)) * 100 /
SUM(total_amount_to_pay),
        2) AS profit_margin
FROM
    fact_orders_enriched
GROUP BY product_name , product_Category
HAVING Revenue_in_lakhs <=7
    AND profit_margin <=15
ORDER BY Revenue_in_lakhs Asc ;
```

Result 

product_name	product_Category	Revenue_in_lakhs	profit_margin
laborum saepe	Grocery & Staples	0.48	12.39
dicta vero	Grocery & Staples	0.86	14.07
laudantium autem	Grocery & Staples	1.83	9.77
placeat eveniet	Grocery & Staples	2.31	11.68
reprehenderit molestias	Grocery & Staples	2.82	12.33
deleniti autem	Grocery & Staples	3.59	10.59
eos fuga	Grocery & Staples	4.06	11.98
laborum voluptas	Grocery & Staples	4.49	12.79

Insights : -

- In this , products are covering (0.91 %) of total_Revenue
- 100 % products from "Grocery & Staples " Category
- Having an Average Profit_margin of (11.45 %)
- Laudantium autem from "Grocery & Staples " Tops in product in LowSales (1.83 Lakhs) with Low_Profit_margin(9.77 %)

Customer Level insights

Customer Level insights : -

1. KPI's

A. - objective : finding Our Total_customer we have

```
SELECT  
COUNT(DISTINCT (Customer_id)) AS Total_unique_Custome  
FROM  
fact_orders_enriched;
```

Total_unique_Custome
18414

B. -- objective : Analysing "AVG Retention_Rate" Monthly each Year

"Query View"

https://drive.google.com/file/d/1wVDJWahcHI55i1MReAIP25XWB9t-E6gD/view?usp=drive_link

AVg_Retention_Percentage_2023	AVg_Retention_Percentage_2024
27.39	62.23

C. -- objective : Analysing "AVG churned_Rate" Monthly each Year

"Query View"

https://drive.google.com/file/d/1AzFNCI5beB_pmW798YTpjHnHEajl-fQy/view?usp=drive_link

AVg_Churned_Percentage_2023	AVg_Churned_Percentage_2024
72.61	37.77

D. -- objective : Analysing "AVG_#_New_Customer_joining" Monthly each Year

"Query View"

https://drive.google.com/file/d/16welaXivkVJDRIdnbXlggnwsekD5Vb6s/view?usp=drive_link

AVG_#_New_Customer_2023	AVG_#_New_Customer_2024
1012.08	522.42

E. -- objective : finding Average Customer_Growth_Rate_Monthly each Year

"Query View"

https://drive.google.com/file/d/1i_IEQtLgAPhyq38wM7B2C2WXZmjljOAC/view?usp=drive_link

AVG_#_New_Customer_2023	AVG_#_New_Customer_2024
1012.08	522.42

2. -- objective : Analyzing which one drives the more revenue "Repeat Customer or New_Customer " in Each Month for Each Year

```

with c1 as (
SELECT
customer_id, month(MIN(order_Date)) AS first_purchase_month ,
year ( MIN(order_Date) ) as first_purchase_Year
FROM
fact_orders_enriched
GROUP BY customer_id
),

c2 as (
SELECT
year,
customer_id,
order_date,
total_Amount_to_Pay,
first_purchase_month,
first_purchase_Year
FROM
fact_orders_enriched
JOIN
c1 USING (customer_id)
),

c3 as (
SELECT
*,
CASE
WHEN month(order_Date) = first_purchase_month and year(order_Date) = first_purchase_Year THEN 'New_Customer'
ELSE 'Repeat_Customer'
END AS R_N_Customer
FROM
c2
)
SELECT
year,
ROUND(SUM(CASE
WHEN R_N_Customer = 'Repeat_Customer' THEN total_Amount_to_Pay
END) / 1000000,
2) AS Rev_Generated_with_RpT_Cust_Mill,
ROUND(SUM(CASE
WHEN R_N_Customer = 'New_Customer' THEN total_Amount_to_Pay
END) / 1000000,
2) AS Rev_Generated_with_New_Cust_Mill
FROM
c3
GROUP BY year;

```

year	Rev_Generated_with_RpT_Cust_Mill	Rev_Generated_with_New_Cust_Mill
2023	27.67	83.61
2024	68.10	42.86

Insights : -

In 2023 - (73 %) of Revenue Was driven by New_Customer and Only (27 %) was Driven By Repeat Customer (Retention Strategies did not work)

In 2024 - (63 %) of Revenue was derived from Repeat Customers and (37 %) of Revenue from New Customers (Retention Strategies Have improved Well by (+ 36 %) From last year

3.-- objective : Doing the Pareto Analysis(80/20) with customers to see the Contribution Spread ‘

```
with c1 as (  
  SELECT  
  customer_id,  
  ROUND(SUM(total_Amount_to_pay) / 1000, 2) as Total_spending_k,  
  COUNT(DISTINCT (order_id)) AS Total_orderd_yet,  
  ROUND((SUM(total_Amount_to_pay) / COUNT(DISTINCT (order_id))) / 1000,  
    2) AS AOV_k  
FROM  
  fact_orders_enriched  
  GROUP BY customer_id  
ORDER BY Total_spending_k DESC  
,  
c2 as (  
  SELECT  
    *,  
    ROUND((Total_spending_k * 100) / (SELECT  
      ROUND(SUM(total_Amount_to_pay) / 1000, 2)  
    FROM  
      fact_orders_enriched),  
    3) AS contribution_to_Overall_sales  
FROM  
  c1  
)  
,  
c3 as (  
  select *,  
  sum(contribution_to_Overall_sales) over(order by Total_spending_k desc) as rolling__sum_of_contribution  
from c2  
)  
select * from c3 where rolling__sum_of_contribution<=80;  
  
select round(9950*100/18414,2) as "%_of_customer_covering_80%_of_REv"; -- to check the percentage of  
Customoers
```

Result 

	%_of_customer_covering_80%_of_REv
►	54.03

4.-- objective : Doing "R F M" (R - Recency , F- Frequency , M-Monetary) and Adding " Customer Segments " to Analyse

```
with c1 as (
SELECT
    customer_id,
    datediff((select max(order_Date) from fact_orders_enriched),max(order_date)) as
Days_since_Last_Purchase,

    ROUND(SUM(total_Amount_to_pay) / 1000, 2) AS Total_spending_k,
    COUNT(DISTINCT (order_id)) AS Total_orderd_yet,
    ROUND((SUM(total_Amount_to_pay) / COUNT(DISTINCT (order_id))) / 1000,
    2) AS AOV_K
FROM
    fact_orders_enriched
GROUP BY customer_id
ORDER BY Total_spending_k DESC
),

c2 as (
    select *,
    ntile(5) over ( order by Total_spending_k desc ) as Table_divider
from c1
),

c3 as (
select *,
case when Table_divider = 1 then "High spender"
    WHEN Table_divider IN (2,3,4) THEN "Mid level Spender "
    else " Low Spender" end as Spend_Group ,

    case when Days_since_Last_Purchase <=30 then "Active"
    WHEN Days_since_Last_Purchase >=31 and Days_since_Last_Purchase <=60 THEN "At Risk"
    else "Churned" end as customer_Status
from c2
)

SELECT
    customer_id, Days_since_Last_Purchase, Total_spending_k, Total_orderd_yet, AOV_K,
    Spend_Group, customer_Status
FROM
    c3 ;
```

“ SQL Query Result”

https://drive.google.com/file/d/1_oqxwfmfXDqWsrh9E4EVBQCxbDAjqap/view?usp=drive_link

Insights :

- 1. (20 %) of Customer are High_spending Customer Covering (42.20 %) of Total_Revenue Generated (in 20 % , 15.5 % (churned) , 2.10 % (at Risk of churn)**
- 2. (60 %) of customer are Mid_Level_Spender Covering (52.42 %) of Total_Revenue Generated (in 60 % , 52.5 % (churned) , 3.73 % (at Risk of Churn)**
- 3. (20 %) of Cust are low_spenders (in 20 % , 18.06 % (churned) , 1 % at Risk of Churn) Generates 4% Rev**

5-- objective : Analysing ("Churned" and "At Risk") Customer's Revenue Contribution OR _At Risk Amount_ -

```
with c1 as (
SELECT
    customer_id,
    DATEDIFF((SELECT
                MAX(order_Date)
            FROM
                fact_orders_enriched),
            MAX(order_date)) AS Days_since_Last_Purchase,
    ROUND(SUM(total_Amount_to_pay) / 1000, 2) AS Total_spending_k,
    COUNT(DISTINCT (order_id)) AS Total_orderd_yet,
    ROUND((SUM(total_Amount_to_pay) / COUNT(DISTINCT (order_id))) / 1000,
    2) AS AOV_K
FROM
    fact_orders_enriched
GROUP BY customer_id
ORDER BY Total_spending_k DESC
),

c2 as (
    select *,
    ntile(5) over ( order by Total_spending_k desc )
as Table_divider
from c1
),

c3 as (
select *,
case when Table_divider = 1 then "High spender"
      WHEN Table_divider IN (2,3,4) THEN "Mid level Spender "
      else " Low Spender" end as Spend_Group ,

      case when Days_since_Last_Purchase <=30 then "Active"
      WHEN Days_since_Last_Purchase >=31 and Days_since_Last_Purchase <=60 THEN "At Risk"
      else "Churned" end as customer_Status
from c2
),

c4 as (
    select count(distinct(customer_id)) from c3
where Spend_Group ="High spender"
)

select
customer_Status,
count(distinct(customer_id)) as Total_Customers_Are,
round((sum(Total_spending_k)*1000)/10000000,2) as Total_Amount_At_Risk_CR,

round(((sum(Total_spending_k)*1000)/10000000)*100/(select sum(total_amount_to_pay)/10000000 from
fact_orders_enriched) ,2) as percentage_of_Overall_Sales,

count(distinct(( case when Spend_Group = "High spender" then customer_id end ))) as No_of_high_spend_Customer,

round(count(distinct(( case when Spend_Group = "High spender" then customer_id end )))*100/(select * from c4 ),2) as
percentage_of_total_high_spender
from c3
where
e customer_Status = "Churned" or customer_Status = "At Risk"
group by customer_Status ;
```

Insights : -

customer_Status	Total_Customers_Are	Total_Amount_At_Risk_CR	percentage_of_Overall_Sales	No_of_high_spend_Customer	percentage_of_total_high_spender
At Risk	1243	1.85	8.33	388	10.53
Churned	15844	18.38	82.69	2854	77.49

6.-- objective : Showing Top 20 customers in terms of Total_spending

```
with c1 as (
SELECT
  customer_id,
  DATEDIFF((SELECT
    MAX(order_Date)
  FROM
    fact_orders_enriched),
    MAX(order_date)) AS Days_since_Last_Purchase,
  ROUND(SUM(total_Amount_to_pay) / 1000, 2) AS Total_spending_k,
  COUNT(DISTINCT (order_id)) AS Total_orderd_yet,
  ROUND((SUM(total_Amount_to_pay) / COUNT(DISTINCT (order_id))) / 1000,
    2) AS AOV_K
FROM
  fact_orders_enriched
GROUP BY customer_id
ORDER BY Total_spending_k DESC
),
c2 as (
  select *,
  ntile(5) over ( order by Total_spending_k desc ) as Table_divider
from c1
),
c3 as (
select *,
case when Table_divider = 1 then "High spender"
      WHEN Table_divider IN (2,3,4) THEN "Mid level Spender "
      else " Low Spender"          end as Spend_Group ,

case when Days_since_Last_Purchase <=30 then "Active"
      WHEN Days_since_Last_Purchase >=31 and Days_since_Last_Purchase <=60 THEN
"At Risk"
      else "Churned" end as customer_Status
from c2
)

select * from c3 order by Total_spending_k desc limit 20;
```

Result ↓

customer_id	Days_since_Last_Purchase	Total_spending_k	Total_orderd_yet	AOV_K	Table_divider	Spend_Group	customer_Status
92725578	194	74.34	7	10.62	1	High spender	Churned
98283663	202	63.87	7	9.12	1	High spender	Churned
50525706	188	63.43	6	10.57	1	High spender	Churned
32886540	464	58.46	5	11.69	1	High spender	Churned
38180225	68	57.88	6	9.65	1	High spender	Churned
20597933	16	57.16	6	9.53	1	High spender	Active
85905355	147	56.98	7	8.14	1	High spender	Churned
8338131	276	56.26	5	11.25	1	High spender	Churned
85122513	67	56.13	7	8.02	1	High spender	Churned
44721373	56	55.79	5	11.16	1	High spender	At Risk
43071980	38	55.66	5	11.13	1	High spender	At Risk
19105476	65	55.17	5	11.03	1	High spender	Churned
41762491	169	55.12	5	11.02	1	High spender	Churned
22633233	191	54.88	5	10.98	1	High spender	Churned
45412348	0	54.41	6	9.07	1	High spender	Active
58572184	48	54.26	6	9.04	1	High spender	At Risk
41515934	244	54.01	6	9.00	1	High spender	Churned
55699471	157	53.92	5	10.78	1	High spender	Churned
91139621	172	53.70	5	10.74	1	High spender	Churned
44162136	126	53.67	5	10.73	1	High spender	Churned

Insights : -

- Top 20 , generates only 0.92 % of Total_Revenue
- In Top 20 , (75 % of customers have Churned) and Rest are (At Risk of churn) except 2 customers
- In Top 20 , (60 %) of Customers Have Churned Long Ago
- AVg spending is (54.2 K)

-- objective : showing Bottom 20 customers in terms of total_spending

```

with c1 as (
SELECT
customer_id,
DATEDIFF((SELECT
MAX(order_Date)
FROM
fact_orders_enriched),
MAX(order_date)) AS Days_since_Last_Purchase,
ROUND(SUM(total_Amount_to_pay) / 1000, 2) AS Total_spending_k,
COUNT(DISTINCT (order_id)) AS Total_orderd_yet,
ROUND((SUM(total_Amount_to_pay) / COUNT(DISTINCT (order_id))) / 1000,
2) AS AOV_K
FROM
fact_orders_enriched
GROUP BY customer_id
ORDER BY Total_spending_k DESC
),

c2 as (
select *,
ntile(5) over ( order by Total_spending_k desc ) as Table_divider
from c1
),

c3 as (
select *,
case when Table_divider = 1 then "High spender"
      WHEN Table_divider IN (2,3,4) THEN "Mid level Spender "
      else " Low Spender" end as Spend_Group ,

      case when Days_since_Last_Purchase <=30 then "Active"
            WHEN Days_since_Last_Purchase >=31 and Days_since_Last_Purchase <=60 THEN "At Risk"
            else "Churned" end as customer_Status
      from c2
      )
select * from c3 order by Total_spending_k limit 20;

```

“ Query Result View  ”

Insights : -

1. In Bottom 20 , Each Customers
Has churned a Long Ago

2. AVg spending (200 rs)

3.AVG no_of_Order (1)

customer_id	Days_since_Last_Purchase	Total_spending_k	Total_orderd_yet	AOV_K	Table_divider	Spend_Group	customer_Status
776572	548	0.03	1	0.03	5	Low spender	Churned
60009074	589	0.03	1	0.03	5	Low spender	Churned
23216381	284	0.03	1	0.03	5	Low spender	Churned
87656659	104	0.03	1	0.03	5	Low spender	Churned
86114894	326	0.04	1	0.04	5	Low spender	Churned
46618101	425	0.05	1	0.05	5	Low spender	Churned
7128753	729	0.05	1	0.05	5	Low spender	Churned
34613284	329	0.05	1	0.05	5	Low spender	Churned
94569866	136	0.05	1	0.05	5	Low spender	Churned
48371128	647	0.05	1	0.05	5	Low spender	Churned
73101300	710	0.06	1	0.06	5	Low spender	Churned
72532429	67	0.06	1	0.06	5	Low spender	Churned
25709218	296	0.06	1	0.06	5	Low spender	Churned
18472243	185	0.06	1	0.06	5	Low spender	Churned
43285595	504	0.06	1	0.06	5	Low spender	Churned
57904018	203	0.06	1	0.06	5	Low spender	Churned
80682998	144	0.07	1	0.07	5	Low spender	Churned
41776288	529	0.07	1	0.07	5	Low spender	Churned
38674259	154	0.07	1	0.07	5	Low spender	Churned
21701244	186	0.07	1	0.07	5	Low spender	Churned

Profitability & Profitable Customer Analysis

1.-- objective : Doing " Peroto Analysis " on the basis of Profit Generated to know How Much percentage of Customer covering 80% of Profit

```
with c1 as (  
  select  
    customer_id ,  
    round((sum(total_Amount_to_pay) - sum(total_Cost_price))/1000,2) as Total_Profit_Acheived_,  
    round( ( round((sum(total_Amount_to_pay) - sum(total_Cost_price))/1000,2))*100/  
    (select round((sum(total_Amount_to_pay) - sum(total_Cost_price))/1000,2) from  
fact_orders_enriched),3)  
  as contribution_to_overall_Profit  
  from fact_orders_enriched  
  group by customer_id  
)
```

```
C2 AS (  
  select *, sum(contribution_to_overall_Profit) over (order by Total_Profit_Archived_ desc)  
  as rolling_contribution_till_80  
  from c1  
)
```

select * from C2 where rolling_contribution_till_80 <=80; – to see the Customers id

select round(7887*100/18414 ,2); – to know the percentage

Result 

PCT_Of_Customer_coveering_80_PCT
42.83

2. Objective : Customer Level Profitability and Spending Analysis Together for deep insights

```
with c1 as (
SELECT
  customer_id,
  datediff((select max(order_Date) from fact_orders_enriched),max(order_date)) as
Days_since_Last_Purchase,
  ROUND(SUM(total_Amount_to_pay) / 1000, 2) AS Total_spending_k,
  ROUND((SUM(total_Amount_to_pay) / 1000 - SUM(total_Cost_price) / 1000),
  2) AS Total_profit_K,
  COUNT(DISTINCT (order_id)) AS Total_orderd_yet,
  ROUND((SUM(total_Amount_to_pay) / COUNT(DISTINCT (order_id))) / 1000,
  2) AS AOV_K
FROM
  fact_orders_enriched
GROUP BY customer_id
ORDER BY Total_spending_k DESC
),
c2 as (
  select *,
  ntile(5) over ( order by Total_spending_k desc ) as Table_divider,
  ntile(10) over(order by Total_profit_K desc) as Profit_Level_Distributer
from c1
),
c3 as (
select *,
case when Table_divider = 1 then "High spender"
      WHEN Table_divider IN (2,3,4) THEN "Mid level Spender "
      else " Low Spender" end as Spend_Group ,

      case when Days_since_Last_Purchase <=30 then "Active"
            WHEN Days_since_Last_Purchase >=31 and Days_since_Last_Purchase <=60 THEN "At Risk"
            else "Churned" end as customer_Status,

      case when Total_profit_K >0 and Profit_Level_Distributer in (1,2) then "High_Profitable_Customer"
            when Total_profit_K > 0 and Profit_Level_Distributer in (3,4,5) then "Moderate_Profit_Customer"
            when Total_profit_K > 0 and Profit_Level_Distributer in (6,7,8) then "Basic_Profitable_Customer"
            when Total_profit_K > 0 and Profit_Level_Distributer in (9,10) then "Very_Low_Profitable_Customers"
            else "Loss_Making_Customer" end as Profit_Group
      from c2
    )
select * from c3;
```

“Query Result View “

https://drive.google.com/file/d/19UrBOxwa9jUFtWK5j5MfLDutQgKqzth8/view?usp=drive_link

Insights : -

1. 20 % of Customers are “ highly_Profitable Customers with the AVG Profit of (9.7 K)
“ (in 20 % , 14.4 % Customer are (High_spender) ,And rest is (Mid_Level_Spender)
2. (30 %) are "Moderate_Profit_Customer" , (30 %) are "Basic_Profitable_Customer" then
(10.4 %) are “ Very_Low_Profitable" and and rest are “ Loss Making _customers”

3. -- Objective : Analyzing Customer's "Spending and Profitability " Together with "Churn"

```
with c1 as (
SELECT
customer_id,
datediff((select max(order_Date) from fact_orders_enriched),max(order_date)) as
Days_since_Last_Purchase,

ROUND(SUM(total_Amount_to_pay) / 1000, 2) AS Total_spending_k,
ROUND((SUM(total_Amount_to_pay) / 1000 - SUM(total_Cost_price) / 1000),
2) AS Total_profit_K,

COUNT(DISTINCT (order_id)) AS Total_orderd_yet,
ROUND((SUM(total_Amount_to_pay) / COUNT(DISTINCT (order_id))) / 1000,
2) AS AOV_K
FROM
fact_orders_enriched
GROUP BY customer_id
ORDER BY Total_spending_k DESC
),

c2 as (
select *,
ntile(5) over ( order by Total_spending_k desc ) as Table_divider,
ntile(10) over(order by Total_profit_K desc) as Profit_Level_Distributer
from c1
),

c3 as (
select *,
case when Table_divider = 1 then "High spender"
      WHEN Table_divider IN (2,3,4) THEN "Mid level Spender "
      else " Low Spender" end as Spend_Group ,

case when Days_since_Last_Purchase <=30 then "Active"
      WHEN Days_since_Last_Purchase >=31 and Days_since_Last_Purchase <=60 THEN "At Risk"
      else "Churned" end as customer_Status,

case when Total_profit_K >0 and Profit_Level_Distributer in (1,2) then "High_Profitable_Customer"
when Total_profit_K > 0 and Profit_Level_Distributer in (3,4,5) then "Moderate_Profit_Customer"
when Total_profit_K > 0 and Profit_Level_Distributer in (6,7,8) then "Basic_Profitable_Customer"
when Total_profit_K > 0 and Profit_Level_Distributer in (9,10) then "Very_Low_Profitable_Customers"
else "Loss_Making_Customer" end as Profit_Group
from c2
)

select count(*) as Total_Customer ,
count( case when Spend_Group ="High spender" then 1 end ) as Total_high_spenders,

count( case when Spend_Group ="High spender" and Profit_Group = "High_Profitable_Customer" then 1
end ) as Custome_high_spender_with_highly_profitable,
count( case when Spend_Group ="High spender" and Profit_Group = "High_Profitable_Customer" and
customer_Status = "Churned" then 1 end ) as Cust_high_spender_with_highly_profitable_and_churned,
```

```

count( case when Spend_Group ="High spender" and Profit_Group = "High_Profitable_Customer" and
customer_Status = "At Risk" then 1 end ) as Cust_high_spender_with_highly_profitable_and_AtRisk,

(sum( case when Spend_Group ="High spender" and Profit_Group = "High_Profitable_Customer" and
customer_Status = "Churned" then Total_spending_k end ) +

sum( case when Spend_Group ="High spender" and Profit_Group = "High_Profitable_Customer" and
customer_Status = "At Risk" then Total_spending_k end ))
as Total_amount_At_Risk_to_loose
from c3;

```

Result 

Total_Customer	Total_high_spenders	Custome_high_spender_with_highly_profitable	Cust_high_spender_with_highly_profitable_and_churned	Cust_high_spender_with_highly_profitable_and_AtRisk	Total_amount_At_Risk
18414	3683	2651	2041	299	62575.47

Insights : -

(20 %) are High_spenders ,

- IN 20 % : -
- 14.4 % are Highly_profitable
- 11.4 % (CustomerWith_highSpend_and_highly_Profitable) have Churned
- 1.6 % ((CustomerWith_highSpend_and_highly_Profitable) Are at Risk of _Churn

